

Explainable Financial RAG: Retrieval Attribution and Faithfulness-Verified Explanations for Financial Question Answering — A Literature Review

Swathi Sowmya Bavirithi
(swathirao1254@gmail.com)

Y. Rama Devi
(yramadevi_cseaiml@cbit.ac.in)

P.Vedh Vishnu
(vedhvishnupogakula@gmail.com)

Parin Uday
(parinuday@gmail.com)

* Department of CSE(AI&ML), Chaitanya Bharathi Institute of Technology, Hyderabad, Telangana, India

Abstract

There are two trust gaps in retrieval-augmented generation (RAG) systems which hinder their adoption in answering natural language questions about financial filings, earnings reports, and tables: (i) retrieval opacity, as the practitioners cannot understand why certain passages were retrieved by the retriever over others during retrieval; and (ii) metric opacity, as the faithfulness scores such as RAGAS of automatically generated answers using retrieved contexts are rarely validated on the perturbations of the retrieved contexts used to generate the answers. This review provides an overview of the literature addressing the aforementioned trust gaps. In particular, the review covers financial question-answering benchmarks (FinQA, TAT-QA), the explainable information retrieval literature and its application to RAG (DeepSHAP-for-retrieval, Rank-LIME, RankingSHAP), the RAG faithfulness-evaluation literature (RAGAS, RAGBench, ARES), and the literature on XAI for financial RAG published in 2025-2026. The review argues that the token/entity-level retrieval attribution for financial questions-answers and the systematic evaluation of RAG faithfulness in finance on perturbations of the retrieved contexts are understudied and can serve as promising research opportunities.

Keywords: Retrieval-Augmented Generation, Explainable AI, Financial NLP, Retrieval Attribution, Faithfulness Evaluation, FinQA, TAT-QA, RAGAS

1. Introduction

Financial question answering presents a higher standard of evidentiary rigor than most other natural language processing tasks. If the system responds that a company's revenues rose by a particular percentage, an analyst, auditor, or regulator may be justified in requiring that the

answer be traced to a particular sentence, table cell, or clause in a filed report, rather than trusting the figure solely based on assertions made within text [1], [2]. For this reason, retrieval-augmented generation (RAG) has become the architecture of choice for financial question answering: the retrieved text provides grounding for the language model's answer, offering stronger guarantees than the model's internal parameters alone, which may exhibit hallucinatory behavior when generating quantitative figures. Two trust gaps, however, appear when RAG moves from research prototypes to production-grade systems.

The first is *retrieval opacity*. A dense retriever scores candidate passages by embedding similarity, but that similarity score is itself an opaque scalar: it does not indicate which words, financial entities, dates, or numeric values in the query actually drove the ranking. Two passages with near-identical surface relevance to a question may receive very different retrieval scores for reasons a user cannot inspect. This matters acutely in finance, where a single digit or reporting period can change the correctness of an answer, and where a user auditing a system's behavior needs more than "this passage was ranked highest."

The second is *metric opacity*. As RAG pipelines have proliferated, so has the practice of certifying their output quality with automated, reference-free metrics — most prominently RAGAS's faithfulness score [6]. These metrics are attractive because they scale without requiring human-annotated ground truth, but they are themselves machine-learned or LLM-judged systems, and a growing body of 2024–2026 work has begun to show that they do not always move in the direction the underlying evidence would suggest [7], [8], [14]. A high faithfulness score, in other words, may not mean what a practitioner assumes it means — and in finance, where such scores could plausibly be used to gate whether an answer is shown to a user at all, that unreliability is a direct safety concern, not an academic curiosity.

The stakes of these two gaps are not purely academic. A financial analyst who cannot see why a retriever preferred one passage over another has no way to audit a system's behavior before relying on it in a report; a compliance reviewer who trusts a faithfulness score without knowing how that score was validated may be certifying an answer that is, in fact, unsupported by its cited evidence. Addressing both gaps together — rather than treating retrieval explanation and faithfulness scoring as separate concerns — reflects how a real financial RAG deployment would need to be audited end to end: a user must be able to ask both "why was this evidence surfaced?"

and "how do I know the system's own claim of faithfulness is itself trustworthy?" This review treats the two questions as complementary halves of a single explainability requirement for financial RAG, which is also how the underlying project frames its two core contributions.

The review is structured as follows. Section 2 provides the background context for this review, describing the financial question-answering task as captured by the FinQA and TAT-QA benchmarks and the RAG architectures they have motivated. Section 3 describes the field of explainable information retrieval, focusing on Anand et al.'s survey of the field [9], which serves as the conceptual foundation for this review. Section 4 describes the technical details of retrieval attribution methods proposed in the explainable IR literature, including DeepSHAP-for-retrieval, Rank-LIME, and RankingSHAP. Section 5 shifts the discussion to the evaluation of faithfulness for answers generated by RAG systems, reviewing the reference-free metrics proposed by RAGAS [6] and the counterarguments raised by RAGBench [7], ARES [8], and FaithJudge [14]. Section 6 highlights the small body of prior work on explainable and trustworthy financial RAG, including several papers from 2025-2026 that are similar in goals and timing to this review's contributions. Finally, Section 7 concludes this review by summarizing the key insights and differentiating this review's contributions from the prior work, and Section 8 offers a comprehensive comparison table of the reviewed literature.

2. Background

2.1 Retrieval-Augmented Generation

A standard RAG pipeline for question-answering over a document corpus typically consists of three stages. First, a retriever encodes the query and the passage retrieved from the corpus into a shared embedding space, typically using a bi-encoder model. Then, given the query and the retrieved passages, a generator model, usually a instruction tuned large language model, generates the final answer. Finally, and most importantly for this work, an evaluation module assesses the triple (query, context, answer) on metrics such as relevance and faithfulness. The appeal of the RAG architecture for financial QA lies in its ability to decouple the language modeling capabilities from the factual grounding in the corpus, hence allowing the latter to be audited. However, the same architecture typically makes the retrieval and evaluation steps opaque, which is the starting point for this paper's contributions.

The retriever is typically a bi-encoder, a model architecture that transforms both query and candidate passages into embeddings in a shared vector space. The degree of relevance between the query and a passage is then calculated using the cosine similarity or dot product between their embeddings. Such models are usually efficient at scale since FAISS or other similar libraries can be used for efficient approximate nearest neighbor search. However, there is a significant caveat in applying such models to financial documents: most often, the embedding space is learned on natural language similarity, whereas financial questions often rely on numerical information. As such, two passages may have very similar embeddings if they share the same phrasing but differ slightly in numbers or have different phrasing but the same set of numbers (e.g., "revenue increased by 5% QoQ" vs. "revenue increased by 5% YoY"). The value of being able to audit the retrieval step in the context of financial QA therefore stems precisely from the desire to analyze what aspects of the passage were responsible for its similarity to the query in the eye of the retriever model. This is especially true if the bi-encoder retriever confuses passages with different numbers but similar language, which is a much more interesting failure mode in the context of the financial domain than, say, retrieving passages with different topics.

Figure 1 illustrates this three-stage pipeline end to end and locates the two trust gaps described above at the exact points in the pipeline where they arise.

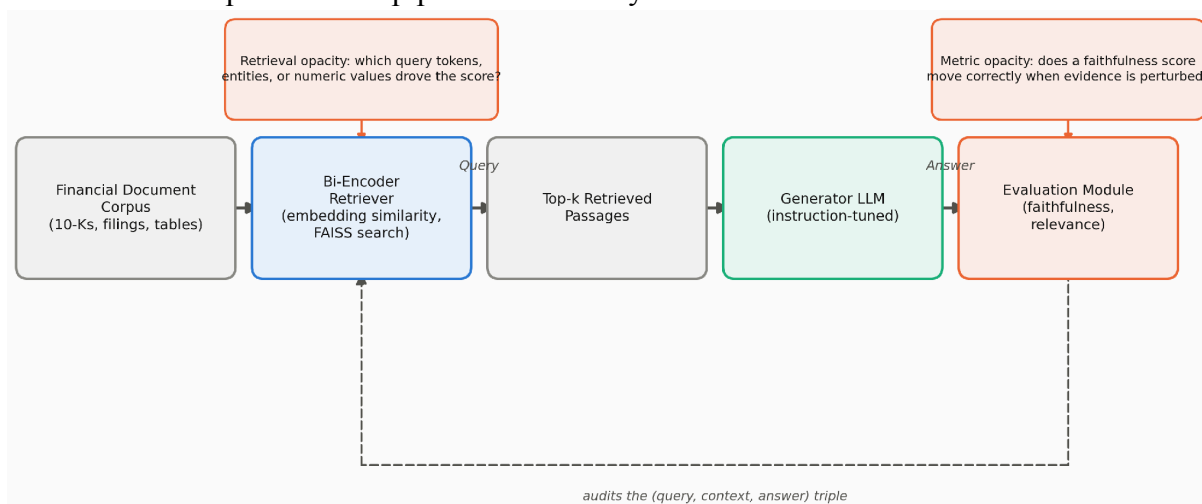


Figure 1. A standard three-stage RAG pipeline for financial question answering, annotated with the two trust gaps this review addresses (Sections 1 and 2.1).

2.2 FinQA and TAT-QA as Financial QA Benchmarks

FinQA [1] and TAT-QA [2] are the two financial question answering benchmarks most widely leveraged as evaluation corpora, and both have been used as such for the project motivating this review. FinQA [1], created by Chen et al. at EMNLP 2021, comprises 8,281 expert-annotated question-answer pairs from S&P 500 earnings reports; each requires multi-step numerical reasoning over narrative text and tables, and -- crucially for their relevance to explainability -- each gold answer is associated with a reasoning program (a sequence of operations such as "subtract" or "divide") over spans of evidence. In other words, FinQA contains built-in reasoning traces for each question, although the systems being evaluated on the benchmark (typically extractive span prediction models) do not always preserve them. TAT-QA [2], by Zhu et al. at ACL 2021, attempts a similar task with 16,552 questions and 2,757 contexts also drawn from financial reports, and introduces a sequence tagging model named TAGOP that identifies relevant spans or table cells prior to applying a symbolic aggregation operator to compute the answer. Both FinQA and TAT-QA present considerable room for improvement over the state-of-the-art (TAGOP scores 58.0 F1 on TAT-QA, compared to a human upper bound of 90.8), which is why these benchmarks continue to be relevant five years after their introduction.

Two aspects of these benchmarks are worth mentioning in relation to the research goal of this review. First, while both FinQA and TAT-QA provide valuable insights into the reasoning capabilities of large language models, they are not RAG-specific, and were created prior to the advent of modern retriever-generator architectures. As such, using these benchmarks to evaluate a complete RAG pipeline (retrieval plus answer generation) requires additional assumptions to be made beyond the ones encoded in the original benchmarks. In particular, both FinQA and TAT-QA are single-document question answering benchmarks, not multi-document. Second, FinQA in particular has been used in downstream research to evaluate RAG systems' ability to retrieve relevant facts with precision, owing to the fact that the correct answer's supporting facts have been manually identified and labeled.

A third consideration, relating this time to the implementation details of the experiments rather than their design, is that the scoring process for the numeric answers provided by the models must be handled with care. Since many of the correct answers in FinQA and TAT-QA are the result of explicit calculations (either ratios or percentage differences between numbers), the answer generation process must be checked for consistency with standard numeric conventions.

In particular, unless steps have been taken to normalize the generated answer (e.g., converting everything to a percentage, removing percentage signs, or converting percentages to decimal form), a correct answer such as "5.2%" will fail to match against "5.2" or "0.052" due to differences in formatting. Any evaluation pipeline using FinQA or TAT-QA answers must therefore make use of a normalization step when comparing the model's answer against the gold standard. Furthermore, a distinction must be made between failures due to incorrect grounding (irrelevant evidence retrieved) and incorrect computation (correct evidence used, but the wrong calculation performed). The difference between the two is crucial to understanding the source of a model's errors, and in particular to evaluating faithfulness, since a standard faithfulness audit may not distinguish between the two types of failures.

3. Explainable AI and Explainable Information Retrieval

Explainability in machine learning has so far focused on feature-attribution methods for classification and regression, such as SHAP, which uses Shapley values from cooperative game theory, and LIME, which approximates the prediction function around an instance by a locally fitted linear model. Both approaches are built for a single-output model, where one can think of the explanation as the contribution of features to a single prediction value. An information retrieval (IR) ranking model, however, produces a list of items as output and considers the relative relevance of all items to the query.

Anand et al.'s 2022 survey, *Explainable Information Retrieval* [9], is the foundational document mapping how the IR community has grappled with this mismatch, and it is the anchor this review uses to frame Section 4. The survey organizes explainable IR (ExIR) work along several axes — post-hoc versus interpretable-by-design methods, local versus global explanations, and the intended stakeholder (end user versus system developer) — and its central diagnostic finding is that most existing ExIR techniques borrow generic XAI machinery (SHAP, LIME, gradient-based saliency) with only partial adaptation to the *listwise*, *contrastive* nature of ranking. That is, a document's relevance in a ranked list is meaningful only relative to the other candidates it was ranked against, and a naive application of SHAP or LIME — designed for single-instance predictions — does not automatically preserve that relative structure. The survey also flags evaluation itself as an open problem in ExIR: because there is rarely a ground-truth label for "this term should have driven this ranking decision," faithfulness and plausibility of

explanations are difficult to measure objectively, a concern that reappears directly in this project's own validation strategy for retrieval attribution (Section 7).

Critically for a financial application, the Anand et al. survey does not itself process financial or numeric text, and it explicitly identifies domain adaptation — handling structured numeric fields, named entities, and multi-token financial terms — as an underserved area within ExIR. This is precisely the adaptation gap the retrieval attribution literature reviewed in Section 4 has begun, but not completed, to close.

Figure 2 summarizes the survey's three organizing axes and the domain-adaptation gap it explicitly leaves open for numeric and entity-dense text such as financial filings.

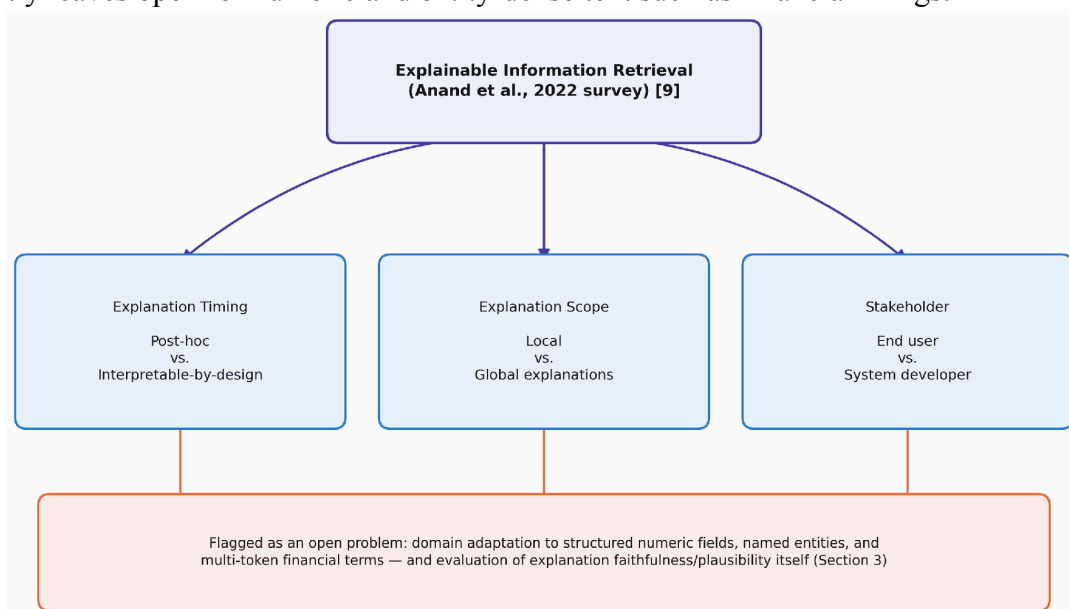


Figure 2. Organizing axes of the Explainable Information Retrieval (ExIR) survey used to frame Section 3, and the domain-adaptation gap it leaves open for financial text.

4. Retrieval Attribution Methods

This section describes the technical predecessors most closely related to the present work's first contribution - determining what query terms, financial entities, dates, or numerical values contributed to a chunk's retrieval score. The paper surveys four methods for this task in depth, and one additional 2025 method that operates at the document level rather than the token level.

4.1 DeepSHAP for Neural Retrieval Models

Fernando, Singh, and Anand's SIGIR 2019 paper [10] is the closest prior art to "applying SHAP to a retriever." DeepSHAP is a hybrid of DeepLIFT and Shapley values that computes term-level attributions by measuring how a term's activation deviates from that of a *reference input*, propagated through the network — a white-box method requiring access to model internals, unlike model-agnostic LIME. The paper's central methodological contribution is also its central finding-as-limitation: because information retrieval has no natural "neutral" baseline analogous to a black image in computer vision, the authors test five different strategies for constructing a reference query-document pair (out-of-vocabulary embeddings, low-IDF stopwords, low query-likelihood words, and two random-document strategies) and find that explanations are highly sensitive to this arbitrary choice, with fewer than half of attributed terms agreeing between DeepSHAP and LIME on the same model. For financial text specifically, this instability is compounded by two further gaps the paper does not address: it operates at the level of single word/term embeddings, with no treatment of multi-token financial entities or numeric values, meaning a dollar figure or percentage would be attributed only as an arbitrary vocabulary token whose behavior depends on tokenization and the (unstable) reference-input choice.

4.2 Rank-LIME

Chowdhury, Rahimi, and Allan's Rank-LIME [11] (SIGIR-ICTIR 2023) adapts LIME specifically to learning-to-rank settings. Where standard LIME perturbs features independently at random, Rank-LIME samples perturbations from a multivariate Gaussian parameterized by the empirical feature covariance observed in training data, keeping perturbed instances closer to the data manifold. Because ranking metrics such as NDCG and Kendall's Tau are non-differentiable, the local surrogate is fit against differentiable ranking-loss proxies (ApproxNDCG performing best among those tested), and the paper introduces new fidelity metrics — Model Fidelity and Explain-NDCG — tailored to list-level, rather than point-level, explanation quality. Evaluated on MS MARCO passage ranking against BM25, fine-tuned BERT, and fine-tuned T5, Rank-LIME improves fidelity by roughly 12% and Explain-NDCG by roughly 18% over baselines, though fidelity for BERT is markedly lower than for BM25 — the more complex and non-linear the underlying ranker, the harder faithful local explanation becomes. For a financial application, two limitations are salient: the method represents documents as unigram bags-of-words, so word order and multi-word entity boundaries ("Federal Reserve," a specific ticker, a quarter label) are

not modeled as units, and numeric values receive no special treatment — under a bag-of-words scheme a specific dollar figure is treated as an arbitrary, likely sparsely observed token, which undermines the covariance-based perturbation strategy the method depends on for realistic sampling.

4.3 RankingSHAP

Heuss, de Rijke, and Anand's RankingSHAP [12] (initially posted to arXiv in 2024, subsequently retitled *RankingSHAP – Faithful Listwise Feature Attribution Explanations for Ranking Models* and published at SIGIR 2025) is the most methodologically rigorous of the attribution papers surveyed here, and the project's stated anchor for its own attribution methodology. Rather than proposing an algorithm first, the paper begins by formally defining what a valid feature attribution *means* for a ranking model — arguing that prior work, including Rank-LIME and DeepSHAP-style approaches, under-specified this — and enumerates axiomatic properties (such as monotonicity and efficiency) an attribution method should satisfy. It then instantiates a *listwise* Shapley-value method: rather than perturbing one document's features in isolation, RankingSHAP applies an identical masking template across every document in the ranked list simultaneously, re-ranks the perturbed list, and scores the resulting change with a listwise objective before aggregating via standard Shapley weighting. This design preserves the contrastive property Section 3 identifies as the central difficulty in adapting SHAP/LIME to ranking. Evaluated on the LETOR 4.0 benchmarks (MQ2008, MSLR-10K) and a synthetic simulation with a known ground-truth importance structure, RankingSHAP recovers the correct feature importances in the synthetic setting where ground truth is available. Its principal limitation for a financial-text application is architectural rather than empirical: RankingSHAP operates over precomputed numeric/categorical query-document feature vectors (the LETOR convention), not raw text — it performs no tokenization or entity handling at all. This is double-edged for a financial retrieval-attribution project: numeric-feature attribution is RankingSHAP's native mode, which is convenient, but applying it to financial *text* retrieval requires first engineering a feature vector (e.g., embedding similarity, extracted numeric fields, entity-match indicators) that the method then explains — the explanation is over those engineered features, not over spans of the original financial passage. A closely related but distinct 2024 paper, RankSHAP (Chowdhury, Zick, and Allan) [21], explicitly critiques

RankingSHAP for not satisfying monotonicity and efficiency axioms and proposes an alternative formalization, underscoring that even the most rigorous end of this literature remains an active, unsettled area rather than a solved problem.

4.4 Document-Level Shapley Attribution in Generation

A 2025 paper by Nematov et al., *Source Attribution in Retrieval-Augmented Generation* [13], extends the Shapley-attribution lineage from the retriever to the generator: it applies Shapley-value attribution to identify which *retrieved documents* most influenced a RAG system's *generated answer*, addressing the computational cost of exact Shapley evaluation when each utility-function call requires an LLM inference, and validating SHAP-style approximations against exact attributions on smaller cases. This paper is the closest 2025–2026 work to the project's first contribution and is treated as a direct point of differentiation in Section 7: it attributes at the document level against the generator's output utility, whereas the project's proposed method attributes at the query token/entity/numeric-value level against the retriever's similarity function — a different granularity and a different target function, applied to a different stage of the pipeline.

4.5 Synthesis

Across Sections 4.1–4.4, a consistent pattern emerges: every method surveyed either (a) operates on generic text tokens with no special handling of numeric values or multi-token financial entities (DeepSHAP, Rank-LIME), or (b) operates on precomputed numeric feature vectors with no handling of raw text at all (RankingSHAP), or (c) attributes at a coarser granularity — whole documents — than query-level tokens or entities (Nematov et al.). None of the four combines token/entity/numeric-level attribution with a target function defined directly over a financial retriever's similarity score, which is the specific technical space the project's Contribution 1 occupies.

Figure 3 traces the chronological lineage of the four attribution methods surveyed above, alongside RankSHAP's critique of RankingSHAP, and marks the gap each leaves for financial text.

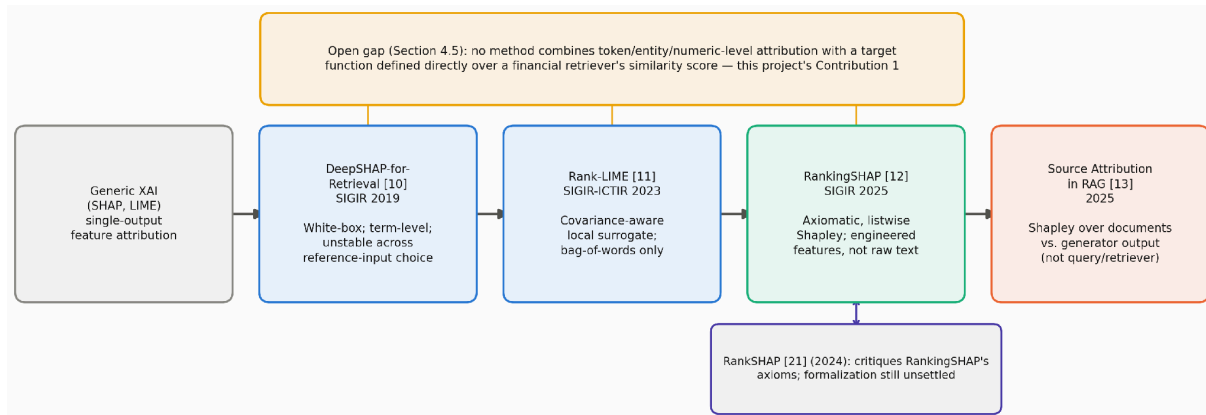


Figure 3. Chronological lineage of retrieval-attribution methods reviewed in Section 4, each method's core limitation for financial text, and the resulting gap this project's Contribution 1 targets.

5. Faithfulness Evaluation of RAG Systems

5.1 RAGAS

RAGAS [6] (Es et al., EACL 2024, first posted to arXiv in September 2023) is the leading reference-free evaluation framework for RAG pipelines, which forms the basis for this project's Contribution 2: stressing the limitations of the "faithfulness" metric proposed by RAGAS. Per the paper, RAGAS evaluates three aspects of a retrieved-answer pair: faithfulness, answer relevance, and context relevance. Faithfulness is calculated based on atomic statements extracted from the answer and checked for "informativeness" - i. e., presence in the retrieved context - by an LLM. Answer relevance is estimated using the cosine similarity between the original question and reverse questions generated from the answer by an LLM, thus measuring focus/verbosity rather than factual accuracy. Context relevance simply measures the ratio of sentences retrieved that are relevant to the question. Of further note - for a related-work section - is that context precision/recall metrics, which sometimes appear in other evaluations attributed to "RAGAS", are not actually present in the original 2023/2024 paper; they were added in later versions of the open-sourced RAGAS library as the tooling continued to develop beyond the original paper's scope, and should therefore perhaps be cited as additions to the software rather than the peer-reviewed article. RAGAS's primary strength is its ability to operate without human

annotations, making it cheap to use in practice, but this same strength is the source of its weakness as demonstrated by subsequent works: using a general-purpose LLM as a judge with minimal prompting leads to imprecisions that specialized methods can exploit.

Figure 4 diagrams how each of RAGAS's three metrics is computed from the underlying (question, context, answer) triple, and where a per-case perturbation audit of the kind Contribution 2 proposes would apply.

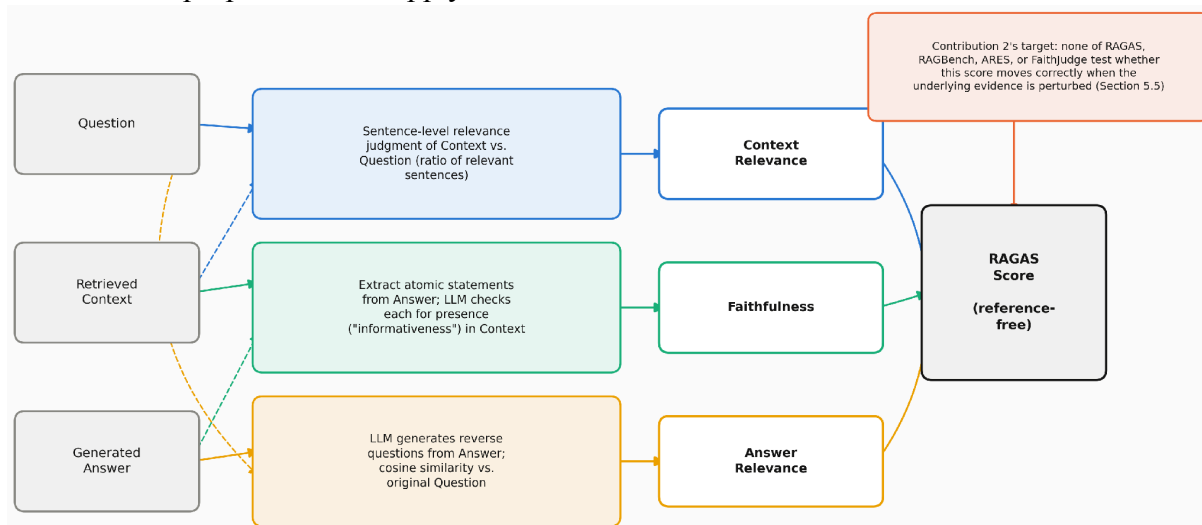


Figure 4. How RAGAS computes its three reference-free metrics from a (question, context, answer) triple (Section 5.1), and the perturbation-testing gap Contribution 2 targets (Section 5.5).

5.2 RAGBench and the Case for Fine-Tuned Judges

RAGBench [7] (Friel et al., arXiv 2024, revised 2025) is a 100,000-example, five-domain benchmark built around the TRACe framework, a set of four metrics — context relevance, context utilization, completeness, and adherence (explicitly equated by the authors with faithfulness/groundedness) — tied to labeled evidence-span annotations rather than an LLM-judged scalar alone. Its central empirical claim, stated directly in the abstract, is that "LLM-based RAG evaluation methods struggle to compete with a finetuned RoBERTa model on the RAG evaluation task": a small, task-specific classifier trained on labeled spans outperforms prompted general-purpose LLM judges of the kind RAGAS uses. This is direct published evidence that prompted-LLM faithfulness scoring — RAGAS's approach — is not the most reliable option available when labeled evidence exists to train against, though it is worth noting RAGBench does not appear to have a confirmed peer-reviewed venue beyond its arXiv listing, and should be cited as a preprint accordingly.

5.3 ARES

ARES [8] (Saad-Falcon, Khattab, Potts, and Zaharia, NAACL-HLT 2024) evaluates RAG systems along the same three RAGAS-style dimensions but replaces zero-/few-shot LLM prompting with lightweight judge models fine-tuned on automatically generated synthetic data for the target domain, and applies prediction-powered inference — using a small set of human-annotated examples (a few hundred) to produce statistically calibrated confidence intervals around the automated scores. Evaluated across eight knowledge-intensive tasks (drawn from KILT, SuperGLUE, and AIS), ARES's fine-tuned-and-calibrated judges outperform RAGAS by an average of approximately 59 percentage points in context-relevance evaluation accuracy and approximately 14 percentage points in answer-relevance accuracy, with per-dataset context-relevance gaps ranging from roughly 48 points (Natural Questions) to as much as 75 points (HotpotQA) — the frequently cited "59-point" figure is the *average* across datasets, not a ceiling, and per-dataset gaps in fact reach higher than that average. ARES also reports higher rank correlation with human judgments than RAGAS (Kendall's tau improvements of 0.065 for context relevance and 0.132 for answer relevance, on average). The cost of this improvement is that ARES requires domain-specific fine-tuning data and several hundred human-annotated calibration examples per target domain — it is not the fully annotation-free approach RAGAS claims to be — but its results constitute strong, directly quantified evidence that an unfine-tuned, uncalibrated LLM judge (RAGAS's default operating mode) can misjudge context relevance by tens of percentage points relative to a better-calibrated alternative.

Figure 5 plots the magnitude of this accuracy gap as reported by Saad-Falcon et al., making clear that the frequently cited 59-point average understates the largest per-dataset gaps.

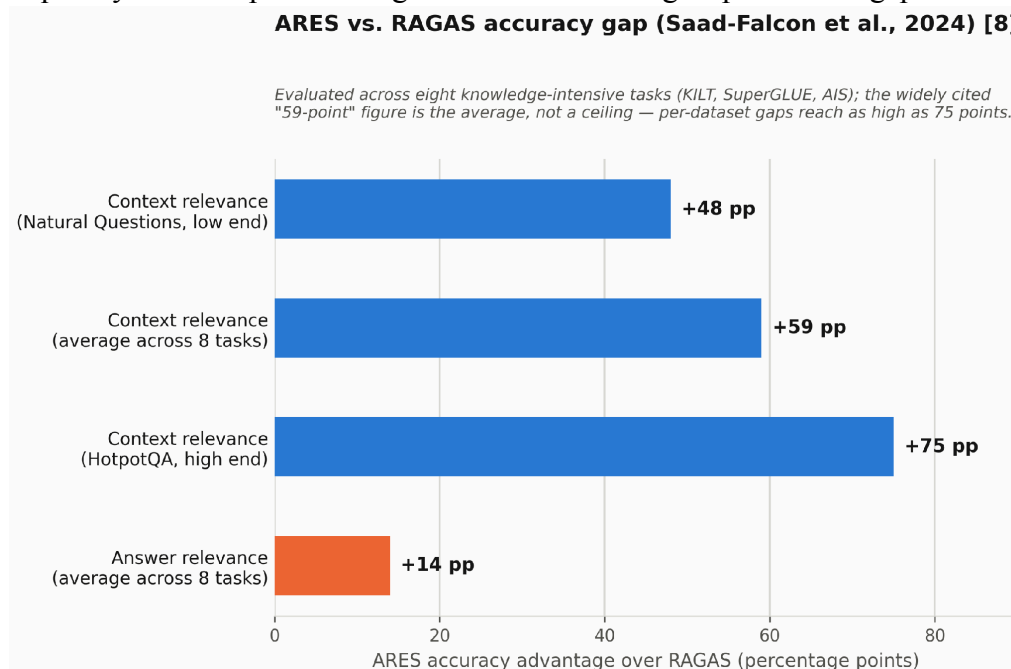


Figure 5. Context- and answer-relevance accuracy gap between ARES's fine-tuned, calibrated judges and RAGAS's zero-/few-shot LLM judge, as reported by Saad-Falcon et al. (Section 5.3).

5.4 Continuing Evidence of Judge Unreliability (2025)

FaithJudge [14] (Tamber et al., EMNLP 2025 Industry Track) provides additional evidence of judge unreliability in the form of an evolving leaderboard; the paper describes the creation of a continuously updated list of hallucination examples for summarization, QA, and data-to-text tasks that can be used to evaluate the performance of various automatic faithfulness judges. The main point of the paper is that models often generate unsupported claims even when given fully relevant context, and that static automated faithfulness judges tend to require frequent recalibration as new examples of hallucination appear, rather than having a single, stable prompting scheme or set of rules. Overall, FaithJudge supports the position that automated faithfulness judging is a non-stationary problem and that the types of judge models explored in RAGAS, RAGBench, and ARES are not immune to changes in performance over time.

5.5 Synthesis

Three independent lines of evidence — RAGBench's fine-tuned-classifier comparison, ARES's calibrated-judge comparison, and FaithJudge's continually-updated human-annotation comparison — converge on the same conclusion: a zero-/few-shot LLM judge of the kind RAGAS deploys by default measurably diverges from human or gold-standard judgments, sometimes by tens of percentage points. What none of these three papers do, however, is test RAGAS's faithfulness score *within a single, fixed RAG pipeline* by directly perturbing or removing the specific evidence a given answer depends on and checking whether the score moves in the corresponding direction on a case-by-case basis. RAGBench, ARES, and FaithJudge each build a *better alternative judge*, evaluated in aggregate against a labeled dataset; none produces a per-case, perturbation-driven reliability audit of RAGAS as it is actually deployed in a specific financial RAG pipeline. That distinction is the opening this review's Section 7 develops into the project's second contribution.

6. Explainable and Trustworthy Financial RAG Systems

This section covers systems that combine RAG with an explicit trust/explanability layer in some form, either directly or indirectly, over financial or closely related high-stakes text.

6.1 Citation- and Attribution-Grounded Financial RAG Systems

Two systems anchor this category: an unreviewed preprint targeting tax-agency documents, and — surfaced only through a dedicated, credibility-filtered search of peer-reviewed venues conducted for this review — the closest fully peer-reviewed competitor identified anywhere in this literature base, a production financial RAG system published at ACL 2026.

Citation-Enforced RAG for Fiscal Document Intelligence [3] is a 2026 pipeline targeting tax-agency document intelligence, circulated as a single-author arXiv preprint with no confirmed peer-reviewed venue as of this writing — a distinction worth noting explicitly, since an unreviewed preprint carries different evidentiary weight than the peer-reviewed sources elsewhere in this review. Its trust mechanism is citation enforcement combined with abstention: the system requires every generated claim to carry a `[Document ID, Page Range]` citation tag (regenerating on failure) and refuses to answer when top-retrieval similarity falls below a threshold. Evaluated on real IRS and state tax documents, it reports 94.5% of claims fully cited and supported, 100% citation-format compliance, 88.0% correct-abstention accuracy, and a 1.8%

hallucination rate. Its authors explicitly scope the system away from numerical computation and personalized advice, limit evaluation to U.S. federal and two state tax authorities, and — most relevant to this review's positioning — build no retrieval-attribution or token/entity-level explanation layer, and no mechanism to validate whether a citation is *actually* the evidentiary basis for a claim beyond format compliance.

The considerably stronger comparator is Katerenchuk, Duboue, and Evanini's *FinRAG-12B: A Production-Validated Recipe for Grounded Question Answering in Banking* [22], published in the ACL 2026 Industry Track — a genuinely peer-reviewed venue, and the most recent, most credible, and most directly on-topic financial-RAG system found anywhere in this review. Developed by Kasisto and deployed live across more than 40 financial institutions, FinRAG-12B LoRA-fine-tunes a Gemma-3-12B-IT model on 98,648 training examples in which every answer is traceable to specific source spans, emits bracketed source-number citations (e.g., "[1]") tied to retrieved passages, and reports 73% citation quality, a calibrated 12% abstention rate, and 3–5x faster and 20–50x cheaper inference than GPT-4.1 at comparable citation quality. Its evaluation includes an explicit "Citation Quality" dimension covering source relevance, citation usage, and a GPT-4.1-judged faithfulness sub-score. This is the single closest peer-reviewed system to this project's overall framing, and it narrows — without closing — the gap the review argues for: its attribution operates at the document/source-number level rather than the query-token/entity level Contribution 1 targets, and its faithfulness sub-score is a static, one-off LLM judgment rather than a perturbation-driven audit of whether that judgment tracks controlled evidence removal, which is what Contribution 2 supplies. Because FinRAG-12B is a proprietary fine-tuning recipe over a 12-billion-parameter production model rather than a published, reproducible attribution or faithfulness-auditing method, it is best treated as this review's primary "nearest deployed system" — the paper a related-work section must most carefully differentiate from — rather than as an architecture the project would directly reproduce and extend in code.

Two further 2025 peer-reviewed papers sharpen this picture. Zhao, Jin, Li, and Gao's *FinRAGBench-V* [23] (EMNLP 2025 main conference) benchmarks multimodal financial RAG in which citations point to visual evidence — charts, tables, page images — rather than text spans, confirming that citation/attribution research in financial RAG is an active main-conference topic, but on a visual-grounding sub-problem distinct from the

query-token/entity attribution this project targets. Wallat, Heuss, de Rijke, and Anand's *Correctness is not Faithfulness in Retrieval Augmented Generation Attributions* [24] (ICTIR 2025) is, read together with the ExIR survey [9], the most important conceptual anchor found for Contribution 2 outside the RAGAS lineage itself: it formally separates citation *correctness* (does the cited passage support the claim) from citation *faithfulness* (did that passage actually cause the generation, versus post-hoc rationalization), and shows that up to 57% of citations in tested systems are unfaithful despite being judged correct. It is general-domain (Natural Questions/Wikipedia), not financial, but it demonstrates that the correctness/faithfulness distinction underlying this project's second contribution is a recognized, live problem in information retrieval generally, rather than a distinction the project invents — the project's contribution is to test that same distinction specifically against RAGAS's automated scoring, under controlled perturbation, in a domain where it has not yet been examined.

6.2 Bayesian RAG (Uncertainty-Aware Financial RAG)

Ngartera, Nadarajah, and Koina's Bayesian RAG [4] (Frontiers in Artificial Intelligence, 2025/2026) proposes a different trust mechanism entirely: a retrieval-scoring function that penalizes epistemic uncertainty, estimated via Monte Carlo Dropout applied to query and document embeddings, alongside mean semantic similarity. Evaluated on real SEC 10-K filings (Apple and Microsoft, FY2023; 1,753 chunks, 24 curated QA pairs), it reports gains over a BM25 baseline (Precision@3 up 20.6%, MRR up 22.7%), a 26.8% reduction in Expected Calibration Error, and a 27.8% reduction in hallucinations. The authors' own stated limitations are notable: the evaluation covers only two companies and 24 QA pairs despite claimed statistical power, adjacent-chunk correlation from overlapping windows is an acknowledged confound, and several documented failure modes — including "semantic similarity traps," where an uncertainty-penalized retriever still surfaces a generic passage over an entity-specific one — are not solved by the uncertainty layer itself. Critically, this system's explainability is a numeric confidence signal attached to a retrieval score, not a source-attribution trail or a claim-level citation; it is complementary to, but architecturally distinct from, both the citation-enforcement approach in Section 6.1 and the token/entity attribution approach this project proposes.

6.3 Reranking and Retrieval-Quality Work Without an Explainability Layer

Cheng et al.'s reranking-analysis paper , also using the FinDER benchmark (drawn from 10-K filings) [5], and Li et al.'s FinGEAR paper (Findings of EMNLP 2025) [17], represent an entirely different approach to financial RAG improvement with relevance to this project's Contribution 1. Both demonstrate large gains in retrieval quality -- Cheng et al. report a cross-encoder reranker boosting an LLM's correctness score from 4.95 to 6.02 ($\uparrow 21.6\%$) across five test groups, while FinGEAR reports substantial gains in F1 over flat vs. graph RAG methods, partially on the FinQA benchmark. Neither includes an explainability or faithfulness verification layer; Cheng et al. explicitly name "explainability features with source attribution" as future work, while FinGEAR's contributions are finance-specific retrieval-quality improvements (guidance, hierarchical indexing) rather than anything to do with metric validation. As such, these works are useful to this project as "state-of-the-art comparison" pieces, and as anchoring points for the argument that retrieval-quality improvements alone do not equal explainability gains.

6.4 Perturbation-Based Robustness and Faithfulness Work Adjacent to Finance

Two papers from 2025-2026 use a perturbation-based methodology similar to this project's Contribution 2, but have significantly different aims and scopes. RARE [15] (Zeng et al., 2025) constructs a pipeline over 527 expert-level finance/economics/policy documents with 48,295 generated questions, applying systematic input perturbations to determine whether RAG answer correctness decreases with changes to the context -- and finding that current RAG systems are generally "unexpectedly sensitive to perturbations", with multi-hop questions being particularly brittle. This is the closest comparable work to this project's Contribution 2, but represents a different problem space: RARE focuses on input perturbations to measure robustness/performance, not on applying input perturbations to train/measure faithfulness of an automated metric like RAGAS. Faithfulness-QA [19] (Ju, Wang, Zhang, 2026) is mechanically the closest comparable work to this project's Contribution 2: it creates entity substitution perturbations to produce knowledge-conflicting context/model parameter pairs and trains context faithfulness detectors on general-domain QA (SQuAD, TriviaQA) with those synthetic examples. It differs from this project in three ways: it is general-domain rather than finance-specific; it focuses on entity substitution rather than evidence removal or top-token masking; and it uses the entity substitution to train faithfulness models rather than measure systematicity of automated metrics like RAGAS. Finally, note that Faithfulness-QA's provenance

is somewhat suspect -- Ju appears to be employed by an AI paper discovery product, not an academic institution; the paper was not presented at a venue -- so that work is explicitly cited as a real but unreviewed preprint.

6.5 Financial Hallucination Benchmarks

PHANTOM [16] (Ji et al., NeurIPS 2025 Datasets and Benchmarks Track) and K-FinHallu [18] (Cho et al., KAIST AI and Financial Tech Lab + KakaoBank, 2026) represent two works investigating financial hallucination detection as a separate problem from general hallucination detection. Both works demonstrate that financial hallucination detection requires substantially different metrics from general-domain hallucination detection -- PHANTOM benchmarking shows the former has failed in numerical precision, financial terminology, and long-context processing, while K-FinHallu's multi-turn Korean financial RAG experiment shows that even the best LLMs are not reliable at diagnosing individual hallucinations, and specifically -- as per Section 6.1 -- are unable to reliably make the "refusal" choice. Neither paper conducts the kind of perturbation experiment this project's Contribution 2 undergoes, testing whether existing automated hallucination/detection metrics can reliably recognize the lack of faithfulness after removal of specific evidence sentences, tokens, or entities; both are static hallucination detection benchmarks, not tools for training or testing specific metrics. As such, they serve as supporting evidence for this project's main thesis: that the financial domain has specific reliability challenges, and general-domain validation of a metric like RAGAS is not sufficient for downstream usage without specialized testing.

6.6 A Contrasting Trust Mechanism: Market-Feedback Adaptive Retrieval

Zhao and Welsch's point-in-time financial RAG system [20] (MIT, 2026) represents a fundamentally different approach to trust generation than any explored in this project. Their system freezes the generator component and updates a Bayesian source memory with realized market-return feedback, allowing it to adapt retrieval strategies to changing market conditions. It demonstrates retrieval adaptation being more effective than reader fine-tuning for its specified event-impact prediction task. Unlike any of the papers reviewed in this section, it does not enforce source citations, conduct explicit uncertainty estimation, or perform any type of faithfulness measurement or detection -- it is included as another example of the variety of

methods falling under the "trustworthy financial RAG" umbrella, with the caveat that none of them directly address the two gaps this project attempts to fill.

6.7 The Financial RAG Trust Landscape

Taken together, these works demonstrate that there is no unified view of what constitutes "trustworthy financial RAG" -- rather, there are multiple different approaches, each targeting a specific aspect of trust generation. Some, like the papers in Section 6.1, attempt to enforce source citations and ensure that if an answer can be attributed to a document, it actually came from that document. Others -- notably the paper in Section 6.2 -- attempt to estimate uncertainty in a probabilistic manner, penalizing retrieval choices that the model itself is not confident about. Section 6.3's papers represent a different school of thought altogether, targeting general-purpose retrieval quality as a proxy for trust rather than targeting explainability, detection, or prevention of unfaithfulness directly. There also exist methods focusing on detection (Sections 6.4-6.5) and different approaches to adaptation (Section 6.6). Not one of these works directly targets the problem this project does -- providing an audit of a claim-level faithfulness metric (Section 5) or implementing attribution mechanisms (Section 4) -- hence why they are discussed collectively in this section, rather than individually.

Figure 6 organizes the systems discussed in this section by the category of trust mechanism each implements, and summarizes why none targets the intersection this project's two contributions occupy.

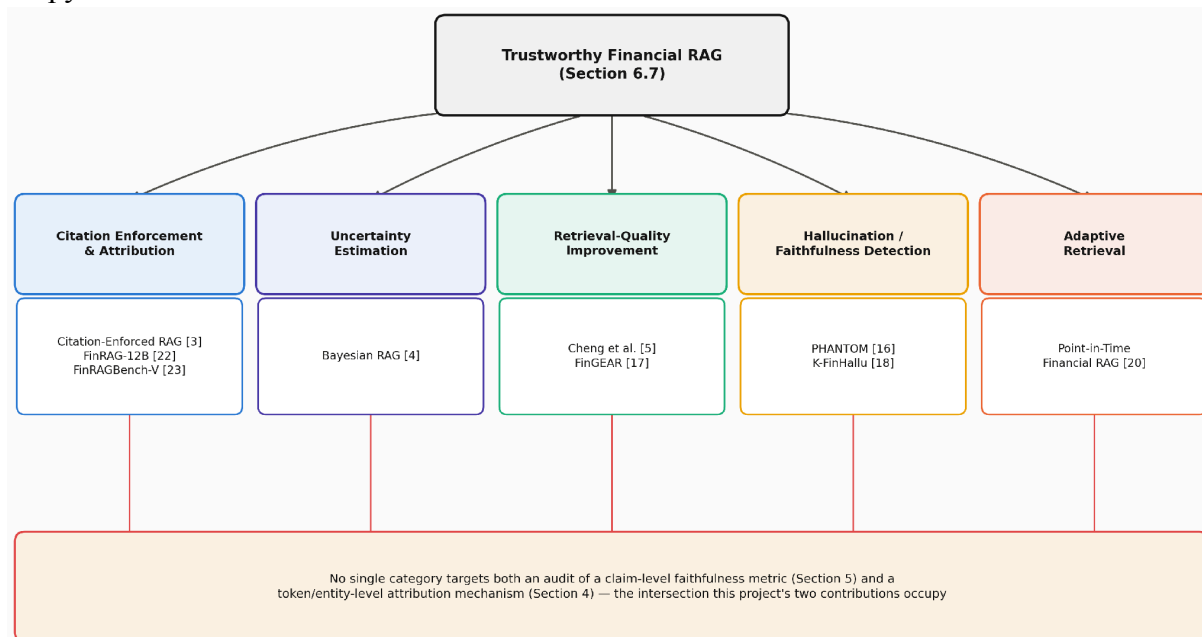


Figure 6. Trust-mechanism taxonomy for the financial RAG systems reviewed in Section 6, grouped by the category of trust each targets.

7. Synthesis and Research Gap

The literature reviewed in Sections 4–6 supports two related but distinct conclusions. First, retrieval attribution *as a technique* is a maturing subfield (Section 4), but every method surveyed leaves an unresolved handling gap for financial text specifically: generic-token methods (DeepSHAP, Rank-LIME) have no numeric- or entity-aware treatment, and the most rigorous listwise method (RankingSHAP) operates on engineered numeric features rather than raw query text, requiring a feature-engineering step this project's proposed method (which perturbs the query directly against the retriever's similarity function) would not need. The 2025 extension to document-level generation attribution (Nematov et al. [13]) confirms the lineage is active but has not moved toward token/entity-level financial retrieval attribution. Second, faithfulness-metric reliability *as a concern* is well established in the general domain (Section 5 — RAGBench, ARES, and FaithJudge all show RAGAS-style scores diverge measurably from better-calibrated alternatives), and finance-specific hallucination is independently established as a distinct problem (Section 6.5 — PHANTOM, K-FinHallu), but no surveyed work combines the two: a

controlled, per-case, perturbation-driven audit of whether RAGAS's faithfulness score moves correctly when the specific evidence a financial answer depends on is removed. RARE [15] comes closest by perturbing inputs in a finance-adjacent domain, but tests answer-correctness robustness, not metric-reliability; Faithfulness-QA [19] comes closest methodologically (perturbation to probe faithfulness) but trains models on general-domain data rather than auditing an existing metric in finance.

Table 1 below highlights the differences for the six most relevant works identified, as well as the two additional competing methods found during the literature search.

Table 1. Differentiation from the closest related work

Paper	Trust/Explainability Mechanism	Domain	What It Does Not Address	Delta From This Project
Citation-Enforced RAG / Auditable LLM-RAG [3]	Citation enforcement + abstention on low retrieval similarity	Tax/fiscal documents	No token/entity-level attribution; no validation that a citation is the actual evidentiary basis beyond format compliance	Targets citation <i>fidelity</i> ; this project targets <i>why</i> a chunk was retrieved and <i>whether</i> the faithfulness score tracking that retrieval is itself reliable
FinRAG-12B [22] (peer-reviewed, ACL 2026 Industry Track; closest overall competitor)	Document/source-number citation tags + calibrated abstention, LoRA-tuned on grounded QA data	Banking (production system, 40+ institutions)	Attribution is document-level, not token/entity-level; faithfulness is a static, one-off LLM-judged sub-score, not a perturbation audit	Project targets finer query-token/entity attribution against retriever similarity, and a perturbation-driven audit of faithfulness-metric <i>reliability</i> rather than a one-off judged score
Wallat et al., Correctness $\neq$ Faithfulness [24] (ICTIR 2025)	Formal separation of citation correctness vs. citation faithfulness	General QA (Natural Questions/ Wikipedia)	Not finance-specific; not tied to RAGAS or any specific automated metric	Project applies this same correctness/faithfulness distinction specifically to RAGAS's automated score, under controlled perturbation, in finance
Bayesian RAG [4]	Monte Carlo Dropout uncertainty penalizing retrieval score	SEC 10-K filings (2 companies)	No source-attribution trail; "semantic similarity traps" persist despite uncertainty penalty	Different trust mechanism (confidence vs. attribution); does not explain <i>which</i> query features drove a score, only how confident the system is in it
Anand et al., ExIR Survey [9]	Taxonomy paper, no method of its own	General IR	Explicitly flags numeric/entity domain adaptation as an open problem	This project is a direct instance of the domain-adaptation gap the survey identifies, applied to finance

Paper	Trust/Explainability Mechanism	Domain	What It Does Not Address	Delta From This Project
DeepSHAP-for-retrieval [10]	White-box Shapley/DeepLIFT hybrid, reference-input dependent	General neural retrieval (TREC Robust04)	No multi-token entity handling; no numeric-value treatment; explanations unstable across reference-input choice	Project needs a numeric/entity-aware, reference-stable attribution approach for financial queries
Rank-LIME [11]	Correlation-aware local surrogate over bag-of-words features	General ranking (MS MARCO)	No word-order/entity-phrase modeling; numeric tokens treated as arbitrary, sparse vocabulary	Financial entities and numeric magnitudes need to be attributed as units, not independent bag-of-words tokens
RankingSHAP [12]	Axiomatic, listwise Shapley attribution	General ranking (LETOR 4.0)	Operates on precomputed numeric/categorical features, not raw text; no tokenization	Project's methodology is anchored here for listwise rigor but must extend it to operate directly over financial query text
Source Attribution in RAG [13]	Shapley attribution over retrieved documents vs. generator output	General RAG	Attributes at document level, against generation utility, not query-token level against retriever similarity	Different granularity (document vs. token/entity) and target function (generator vs. retriever)
RARE [15]	Systematic input perturbation testing answer-correctness robustness	Finance/economics/policy documents	Tests whether <i>answers</i> degrade under perturbation, not whether the <i>faithfulness metric</i> correctly tracks evidence removal	Project audits metric reliability per case, not system robustness in aggregate
Faithfulness-QA [19]	Counterfactual entity-substitution perturbation	General-domain QA (SQuAD/TriQA)	Perturbs to <i>train</i> faithful models; general domain; entity substitution rather than evidence-sentence removal	Project perturbs to <i>audit</i> an existing deployed metric (RAGAS) in financial, numerically dense QA

Two further observations from Section 4.5 and Section 14 of the underlying project brief are pertinent to this discussion. First, since there can be no ground-truth labeling of which tokens "should have driven retrieval," the verification step - testing the claim that higher-attribution tokens, when removed, alter the retrieved ranking more than lower-attribution ones - must be explicitly framed as a hypothesis of necessity/sufficiency rather than a certainty, which aligns this project's evaluation design with the ExIR survey's [9] framing of evaluation as an open problem for the field. Second, by virtue of FinQA's existing gold supporting-fact annotations [1], this project does not have to invent a novel ground-truth metric for retrieval quality; it has the option to align its evaluation either with FinQA's existing metrics or to build upon them.

Figure 7 positions the works discussed above, and this project's two contributions, along the two dimensions the surveyed literature varies on.

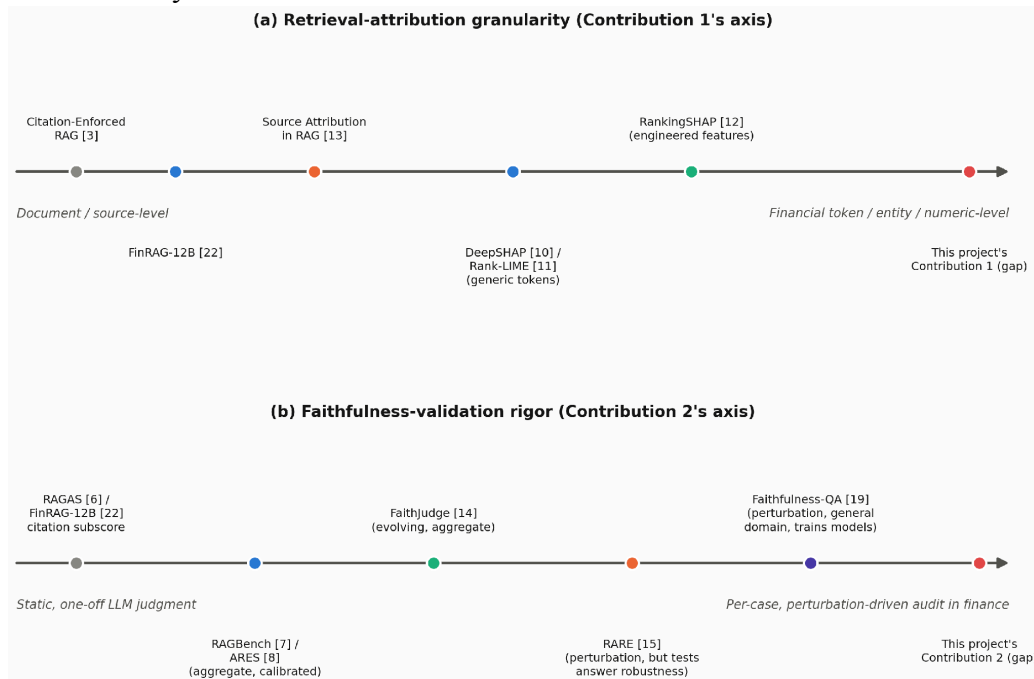


Figure 7. Positioning this review's two research gaps (Sections 7 and 9) along the two dimensions the surveyed literature varies on: retrieval-attribution granularity and faithfulness-validation rigor.

8. Comparative Summary Table

Table 2 summarizes the method, dataset(s), evaluation metrics, and limitation of each paper reviewed in Sections 2-6; together, they form this review's complete literature base.

Table 2. Summary of reviewed literature

#	Paper	Method / Contribution	Dataset(s)	Key Metrics	Key Limitation
[1]	FinQA (Chen et al., 2021)	Numerical-reasoning QA dataset with program annotations	S&P 500 earnings reports (8,281 QA pairs)	Execution accuracy, program accuracy	Static single-document task; not a RAG or faithfulness benchmark
[2]	TAT-QA (Zhu et al., 2021)	Hybrid tabular/textual QA dataset + TAGOP model	Financial reports (16,552 QA pairs)	F1, Exact Match	No multi-document retrieval; predates LLM-based RAG
[3]	Citation-Enforced RAG for Fiscal Document Intelligence	Citation enforcement + abstention	IRS/state tax documents	% claims cited, hallucination rate, abstention accuracy	No attribution layer; single-author arXiv preprint, not peer-reviewed

#	Paper	Method / Contribution	Dataset(s)	Key Metrics	Key Limitation
[4]	Bayesian RAG (Ngartera et al.)	MC Dropout uncertainty-penalized retrieval scoring	SEC 10-K filings (2 companies, 24 QA pairs)	Precision@3, MRR, ECE, hallucination rate	Very small evaluation scale; no source-attribution trail
[5]	Reranking Analysis / FinDER (Cheng et al.)	Hybrid BM25+dense retrieval with cross-encoder reranking	FinDER benchmark (10-K filings, 1,500 queries)	LLM-judge correctness score	No explainability layer (explicitly future work)
[6]	RAGAS (Es et al., 2023/2024)	Reference-free LLM-judged RAG evaluation	General RAG pipelines	Faithfulness, answer relevance, context relevance	Zero-/few-shot judge, no domain calibration
[7]	RAGBench (Friel et al., 2024/2025)	Labeled-span benchmark; fine-tuned classifier judge (TRACe)	5 industry domains, 100k examples	Context relevance, utilization, completeness, adherence	Domain-limited; venue not confirmed peer-reviewed
[8]	ARES (Saad-Falcon et al., 2024)	Fine-tuned judge + prediction-powered inference calibration	KILT, SuperGLUE, AIS (8 tasks)	Context/answer relevance & faithfulness accuracy vs. RAGAS	Requires per-domain fine-tuning data and human calibration set
[9]	Explainable IR Survey (Anand et al., 2022)	Taxonomy of ExIR methods	N/A (survey)	N/A	No empirical method; flags but doesn't solve domain adaptation
[10]	DeepSHAP-for-retrieval (Fernando et al., 2019)	Shapley/DeepLIFT attribution for neural retrievers	TREC Robust04	Attribution overlap across reference strategies	Explanations unstable across reference-input choice
[11]	Rank-LIME (Chowdhury et al., 2023)	Correlation-aware local surrogate for ranking	MS MARCO	Model Fidelity, Explain-NDCG	Bag-of-words; no entity/numeric handling
[12]	RankingSHAP (Heuss et al., 2024/2025)	Axiomatic listwise Shapley attribution	LETOR 4.0 (MQ2008, MSLR-10K)	Recovery of ground-truth importances (synthetic)	Operates on engineered features, not raw text
[13]	Source Attribution in RAG (Nematov et al., 2025)	Shapley attribution over retrieved documents in generation	General RAG	Approximation accuracy vs. exact Shapley	Document-level granularity; generation-side target function
[14]	FaithJudge (Tamber et al., 2025)	Evolving human-annotated hallucination leaderboard	Summarization, QA, data-to-text	Faithfulness judge accuracy vs. human labels	General-domain; requires continual human relabeling
[15]	RARE (Zeng et al., 2025)	Perturbation-based robustness testing	527 finance/econom	Answer-correctness degradation	Tests system robustness, not metric reliability

#	Paper	Method / Contribution	Dataset(s)	Key Metrics	Key Limitation
			ics/policy documents, 48k questions	under perturbation	
[16]	PHANTOM (Ji et al., 2025)	Financial hallucination detection benchmark	Long-context financial QA	Detector accuracy on validated hallucinations	Static benchmark; no metric-reliability audit
[17]	FinGEAR (Li et al., 2025)	Lexicon-guided hierarchical retrieval + reranking	10-K filings incl. FinQA	F1 vs. flat/graph-based RAG	Retrieval-quality only; no explainability component
[18]	K-FinHallu (Cho et al., 2026)	Multi-turn financial hallucination benchmark (Korean)	Real financial documents	Detection/diagnosis accuracy, abstention quality	Different language/market; benchmark, not metric audit
[19]	Faithfulness-QA (Ju et al., 2026)	Counterfactual entity-substitution perturbation dataset	SQuAD/Trivia QA-derived (99,094 examples)	Training-time faithfulness improvement	General domain; trains models rather than audits a metric; unreviewed preprint
[20]	Point-in-Time Financial RAG (Zhao & Welsch, 2026)	Market-feedback adaptive retrieval, frozen LLM	Financial news / SEC filings	Multi-horizon return-prediction accuracy	No explanation surfaced to user; no faithfulness evaluation
[21]	RankSHAP (Chowdhury, Zick & Allan, 2024)	Axiomatic critique and alternative to RankingSHAP	LETOR benchmarks + human study	Monotonicity/efficiency axiom satisfaction	Confirms attribution-for-ranking formalization is still unsettled
[22]	FinRAG-12B (Katerenchuk, Duboue & Evanini, ACL 2026 Industry Track)	LoRA-tuned Gemma-3-12B with citation-tag generation + abstention	Production banking QA (40+ institutions)	Citation quality, abstention rate, cost/latency vs. GPT-4.1	Document-level attribution only; static LLM-judged faithfulness sub-score, no perturbation testing
[23]	FinRAGBench-V (Zhao, Jin, Li & Gao, EMNLP 2025 main)	Multimodal RAG benchmark with visual citation	Bilingual financial reports (~112k pages)	Automatic visual-citation evaluation	Attribution is visual/page-level, not token/entity-level text attribution
[24]	Wallat, Heuss, de Rijke & Anand (ICTIR 2025)	Formal correctness-vs-faithfulness distinction for RAG citations	General QA (Natural Questions/Wikipedia)	% citations faithful vs. correct	General-domain; not tied to RAGAS or any specific automated metric

9. Conclusion

The literature surveyed here supports a specific, bounded novelty claim rather than a broad one. "Explainable financial RAG" as an umbrella is no longer an open space: citation-enforced systems [3], [22], uncertainty-aware retrieval [4], adaptive market-feedback retrieval [20], and a growing set of financial hallucination benchmarks [16], [18] all address some dimension of trust in financial RAG, and reranking-quality and retrieval-quality work [5], [17] continues to push retrieval accuracy forward without an explainability layer. Notably, the single closest peer-reviewed system found in this review — FinRAG-12B [22], a production-deployed banking QA system published at ACL 2026 — already performs document-level citation attribution and a static faithfulness judgment, confirming the field is closing in on this territory but has not yet reached it: its attribution stops at the document level rather than the query-token/entity level, and its faithfulness signal is a one-off judgment rather than a perturbation-validated audit. What remains genuinely underexplored, based on this review, is narrower and more precise: token/entity/numeric-level retrieval attribution specific to financial queries, grounded in the listwise-attribution rigor of RankingSHAP [12] but extended to operate directly on financial query text rather than precomputed features; and systematic, per-case, perturbation-based validation of an automated faithfulness metric (RAGAS) within a single deployed financial RAG pipeline, distinct from both general-domain judge-calibration work [7], [8], [14], the general-domain correctness-versus-faithfulness distinction of Wallat et al. [24], and finance-adjacent but differently-scoped perturbation work [15], [19]. Both gaps sit at the intersection of two otherwise well-populated literatures — explainable information retrieval and RAG faithfulness evaluation — narrowed specifically by the financial domain's numeric density and its correspondingly stricter auditability requirement. This positions the project's two contributions as a direct, well-motivated response to gaps this literature review has identified rather than as speculative extensions of already-crowded territory.

The narrowness of the claim is itself part of the argument. Reviewers of a workshop-track finance-NLP submission are likely to be familiar with RAGAS-style evaluation and with at least one or two explainable-retrieval papers; a novelty claim framed as "the first explainable financial RAG system" would be immediately and correctly challenged given the systems surveyed in Section 6. A novelty claim framed instead as "a system combining financial-query-specific, listwise-attribution-grounded retrieval explanation with a per-case perturbation audit of an existing faithfulness metric in the financial domain" is falsifiable, specific, and — based on the

review above — not yet occupied by existing published work. The accompanying human pilot study planned for this project (a small, explicitly-labeled clarity/trust survey rather than a formal user study) is intended to complement this technical novelty claim with preliminary evidence that the resulting explanations are legible to end users, not to substitute for it.

References

- [1] Z. Chen, W. Chen, C. Smiley, S. Shah, I. Borova, D. Langdon, R. Moussa, M. Beane, T.-H. Huang, B. Routledge, and W. Y. Wang, "FinQA: A dataset of numerical reasoning over financial data," in *Proc. 2021 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2021, pp. 3697–3711. [Online]. Available: <https://arxiv.org/abs/2109.00122>
- [2] F. Zhu, W. Lei, Y. Huang, C. Wang, S. Zhang, J. Lv, F. Feng, and T.-S. Chua, "TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics (ACL-IJCNLP)*, 2021. [Online]. Available: <https://arxiv.org/abs/2105.07624>
- [3] A. C. Shanivendra, "Citation-enforced RAG for fiscal document intelligence: Cited, explainable knowledge retrieval in tax compliance," arXiv:2603.14170, 2026. [Online]. Available: <https://arxiv.org/abs/2603.14170> (unreviewed preprint; no confirmed peer-reviewed venue as of July 2026)
- [4] L. Ngartera, S. Nadarajah, and R. Koina, "Bayesian RAG: Uncertainty-aware retrieval for reliable financial question answering," *Frontiers in Artificial Intelligence*, vol. 8, art. 1668172, 2025/2026, doi: 10.3389/frai.2025.1668172.
- [5] Z. Cheng, L. Lai, Y. Liu, K. Cheng, and X. Qi, "Enhancing financial report question-answering: A retrieval-augmented generation system with reranking analysis," in *Proc. IEEE Int. Conf. Electrical, Computer and Energy Technologies (ICECET)*, Rome, Italy, 2026. [Online]. Available: <https://arxiv.org/abs/2603.16877>
- [6] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated evaluation of retrieval augmented generation," in *Proc. 18th Conf. European Chapter Assoc. Comput. Linguistics: System Demonstrations (EACL)*, 2024, pp. 150–158, doi: 10.18653/v1/2024.eacl-demo.16.
- [7] R. Friel, M. Belyi, and A. Sanyal, "RAGBench: Explainable benchmark for retrieval-augmented generation systems," arXiv:2407.11005, 2024/2025. [Online]. Available: <https://arxiv.org/abs/2407.11005>

- [8] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, "ARES: An automated evaluation framework for retrieval-augmented generation systems," in *Proc. 2024 Conf. North American Chapter Assoc. Comput. Linguistics: Human Language Technologies (NAACL-HLT)*, 2024, pp. 338–354, doi: 10.18653/v1/2024.naacl-long.20.
- [9] A. Anand, L. Lyu, M. Idahl, Y. Wang, J. Wallat, and Z. Zhang, "Explainable information retrieval: A survey," arXiv:2211.02405, 2022. [Online]. Available: <https://arxiv.org/abs/2211.02405>
- [10] Z. T. Fernando, J. Singh, and A. Anand, "A study on the interpretability of neural retrieval models using DeepSHAP," in *Proc. 42nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2019, pp. 1005–1008, doi: 10.1145/3331184.3331312.
- [11] T. Chowdhury, R. Rahimi, and J. Allan, "Rank-LIME: Local model-agnostic feature attribution for learning to rank," in *Proc. 2023 ACM SIGIR Int. Conf. Theory of Information Retrieval (ICTIR)*, 2023, pp. 33–37, doi: 10.1145/3578337.3605138.
- [12] M. Heuss, M. de Rijke, and A. Anand, "RankingSHAP: Faithful listwise feature attribution explanations for ranking models," in *Proc. 48th Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2025, doi: 10.1145/3726302.3729971.
- [13] I. Nematov, T. Kalai, E. Kuzmenko, G. Fugagnoli, D. Sacharidis, K. Hose, and T. Sagi, "Source attribution in retrieval-augmented generation," arXiv:2507.04480, 2025. [Online]. Available: <https://arxiv.org/abs/2507.04480>
- [14] M. S. Tamber, F. S. Bao, C. Xu, G. Luo, S. Kazi, M. Bae, M. Li, O. Mendelevitch, R. Qu, and J. Lin, "Benchmarking LLM faithfulness in RAG with evolving leaderboards," in *Proc. 2025 Conf. Empirical Methods in Natural Language Processing: Industry Track (EMNLP)*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.04847>
- [15] Y. Zeng, T. Cao, D. Wang, X. Zhao, Z. Qiu, M. Ziyadi, T. Wu, and L. Li, "RARE: Retrieval-aware robustness evaluation for retrieval-augmented generation systems," arXiv:2506.00789, 2025. [Online]. Available: <https://arxiv.org/abs/2506.00789>
- [16] L. Ji, D. Seyler, G. Kaur, M. Hegde, K. Dasgupta, and B. Xiang, "PHANTOM: A benchmark for hallucination detection in financial long-context QA," in *Proc. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025. [Online]. Available: <https://openreview.net/forum?id=5YQAo0S3Hm>
- [17] Y. Li, M. Wang, M. de Carvalho, S. Sabanis, and T. Ma, "FinGEAR: Financial mapping-guided enhanced answer retrieval," in *Findings of the Assoc. Comput. Linguistics: EMNLP*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.12042>

- [18] E. Cho, Y. Lee, M. Kim, J. Yang, Y. Kwak, and E. Choi, "K-FinHallu: A hallucination detection benchmark for multi-turn RAG in Korean finance," arXiv:2605.29523, 2026. [Online]. Available: <https://arxiv.org/abs/2605.29523>
- [19] L. Ju, J. Wang, and Q. Zhang, "Faithfulness-QA: A counterfactual entity substitution dataset for training context-faithful RAG models," arXiv:2604.25313, 2026. [Online]. Available: <https://arxiv.org/abs/2604.25313>
- [20] Z. Zhao and R. E. Welsch, "Point-in-time financial RAG with frozen LLMs and market-feedback adaptive retrieval," arXiv:2605.31201, 2026. [Online]. Available: <https://arxiv.org/abs/2605.31201>
- [21] T. Chowdhury, Y. Zick, and J. Allan, "RankSHAP: Shapley value based feature attributions for learning to rank," arXiv:2405.01848, 2024. [Online]. Available: <https://arxiv.org/abs/2405.01848>
- [22] D. Katerenchuk, P. Duboue, and K. Evanini, "FinRAG-12B: A production-validated recipe for grounded question answering in banking," in *Proc. 2026 Conf. Assoc. Comput. Linguistics: Industry Track (ACL)*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.05482>
- [23] S. Zhao, Q. Jin, Y. Li, and X. Gao, "FinRAGBench-V: A benchmark for multimodal RAG with visual citation in the financial domain," in *Proc. 2025 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.17471>
- [24] J. Wallat, M. Heuss, M. de Rijke, and A. Anand, "Correctness is not faithfulness in retrieval augmented generation attributions," in *Proc. 2025 ACM SIGIR Int. Conf. Theory of Information Retrieval (ICTIR)*, 2025, doi: 10.1145/3731120.3744592.